An AI-Enhanced Voice-Enabled Personal Healthcare Assistant with Integrated Caregiver Matching for Chronic Disease Management in Sri Lanka

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***Abstract*—The allocation of home-based caregivers for persons living with non-communicable disease in Sri Lanka continues to rest on informal referral, telephone enquiry, and poorly documented personal networks. Although digital health artefacts have multiplied, published evaluations indicate that store-facing NCD applications remain of only moderate quality, roster systems seldom return a scored and explained shortlist, and conversational agents typically fuse generation with decision in a single model. The present study reports the design, instantiation, and formative evaluation of Care Plus, an AI-enhanced, voice-enabled personal healthcare assistant whose scholarly contribution is an integrated caregiver matcher rather than a diagnostic chatbot. Spoken or typed requests in Sinhala, Tamil, or English are reduced to a constrained care-intent object. A Voice-Enhanced Hybrid Matching Framework then ranks eligible caregivers by a linear fusion of four normalised signals—health compatibility, linguistic and cultural alignment, travel feasibility, and trust—whose coefficients were elicited through the Analytic Hierarchy Process. A stratified purposive instrument of 384 stakeholder records, constructed around the Colombo District role, language, and zone profile, yielded 379 matrices below the consistency threshold of 0.10 (mean individual CR = 0.035; group CR = 0.00005) and the weight vector 0.287, 0.231, 0.280, 0.202. Guided by Design Science Research, the artefact was examined for functionality, ranking behaviour, and usability. All eight critical test cases passed, six of seven walkthrough items were satisfied, and hybrid ranking did not fall below a content-only baseline on the synthetic relevance case. A large language model is confined to dialogue and is excluded from rank determination, thereby preserving explainability and auditability.**

Keywords*—caregiver matching, voice interface, hybrid recommender system, Analytic Hierarchy Process, explainable AI, chronic disease management, design science research, Sri Lanka

I. Introduction

Non-communicable diseases (NCDs)—diabetes, hypertension, and cardiovascular disease among them—account for the greater part of mortality in Sri Lanka and impose a care burden measured in months and years rather than in single clinical episodes [1], [2]. A substantial share of that burden is borne at home. The immediate operational difficulty for many households is therefore not diagnosis but procurement: locating a caregiver whose skills, language, travel radius, and reliability coincide. That search is still conducted through telephone calls, social-media posts, and informal referral. Such channels are slow, they leave the grounds of selection unrecorded, and they afford only weak quality control.

Digital health is frequently advanced as a corrective. Health information technology improves informational flow when it is aligned with the work of care [3], and artificial intelligence has entered both clinical and consumer settings [4]. Store-facing NCD applications, however, continue to exhibit only moderate quality and limited clinical grounding [5], while conversational health agents are rarely examined beyond feasibility [6]. Local conditions tighten the constraint. The recent economic crisis disrupted medicine supply and service continuity [7], and digital health citizenship remains uneven wherever connectivity, literacy, and household budgets are uncertain [8]. In the Colombo District, Sinhala, Tamil, and English are all used in everyday care talk; a form-only interface therefore excludes a non-trivial fraction of the intended users.

The study develops and examines Care Plus, an AI-enhanced, voice-enabled personal healthcare assistant with integrated caregiver matching for chronic disease management. The research object is not a chatbot that assigns caregivers by free-form generation. It is the Voice-Enhanced Hybrid Matching Framework (VEHMF): a numerical ranker that fuses four normalised signals under stakeholder-derived weights and is supplied by structured care intent that may originate as speech. The aim is to establish the practicality and effectiveness of such an assistant when matching and chronic-care management occupy a single pipeline.

A. Research Objectives and Questions

The inquiry was organised around four interrelated objectives. RO1 required a structured review of caregiver-matching platforms in order to isolate empirical, theoretical, and methodological deficiencies in the integration of health monitoring. RO2 required the specification of a hybrid scoring model in which patient need is compared with caregiver capacity under coefficients obtained from stakeholders rather than assigned by the researcher. RO3 required a data-handling design that can be defended against Sri Lanka's Personal Data Protection Act (PDPA), the GDPR, and HIPAA, principally through layered encryption, consent gating, and audit logging. RO4 required an examination of usability and technical behaviour by quantitative checks and structured qualitative judgement.

Each objective is associated with a research question. RQ1 asks by what mechanism AI and machine-learning techniques can raise caregiver–patient matching quality inside a single scoring framework. RQ2 asks to what extent a spoken interface improves access and engagement among users in Colombo who are less confident with digital forms.

RQ3 asks what happens to matching quality when chronic-disease attributes are admitted into the same score. RQ4 asks whether statutory privacy obligations can be observed without rendering the matcher operationally inefficient.

B. Scholarly Contributions

Three contributions arise from the investigation. At the artefact level, the author constructed a voice-to-match pipeline for Sinhala, Tamil, and English care requests in which conversational generation and numerical ranking are institutionally separated. At the empirical level, the study reports a consistency-checked AHP weight vector for caregiver selection under a Colombo stakeholder profile, obtained by the sampling procedure set out in Section III. At the evaluative level, functionality, ranking behaviour, and usability are examined, and claims that were not measured are stated as such. Section II situates the work; Section III records the method, including the sample; Section IV describes the artefact; Section V reports the evaluation; Section VI interprets the results; and Section VII concludes.

II. Literature Review

A. Hybrid Recommender Systems

Caregiver assignment is, computationally, a ranked-retrieval problem over a finite candidate set. The literature distinguishes content-based filtering, which compares item attributes with a user profile, from collaborative filtering, which infers preference from the behaviour of similar users [9], [10]. The former tends to over-specialise; the latter degrades when interaction histories are sparse. Hybrid constructions offset those weaknesses, and among them the weighted hybrid remains unusually transparent [11]. Matrix factorisation is the conventional collaborative baseline [12]; its implicit-feedback variant learns from views and acceptances rather than from explicit ratings [13], a useful property in a marketplace where families seldom rate caregivers. Dense sentence embeddings [14] together with indexes such as FAISS [15] bring content matching within the reach of a modest CPU, drawing on the broader mathematics of similarity and latent factors [16].

In the caregiver setting hybridisation is difficult to avoid. Clinical fit, language and culture, travel burden, and trust pull in different directions; no single similarity score captures all four. That observation is the warrant for a weighted, multi-signal design.

B. Voice Interfaces and Conversational Agents in Health

Spoken interfaces reduce dependence on long forms. Recognition itself has moved from hidden Markov models [17] through deep neural acoustic models [18] to large weakly supervised systems that transcribe many languages [19]. Open corpora such as Common Voice extend resources to under-served languages [20]. Conversational agents have been applied to triage, self-management, and information access, yet most published studies remain small and stop at feasibility [6]. Large language models add fluency, but they also generate statements that are not grounded in their input —a failure mode now termed hallucination [21]. The implication for ranking is direct: if one model both orders the candidates and writes the reason, nothing guarantees that the reason matches the order.

C. Multi-Criteria Weighting with AHP

Where several criteria compete, the Analytic Hierarchy Process derives weights from pairwise judgements on a 1–9 ratio scale [22], [23]. The principal eigenvector of each comparison matrix supplies the weights; the consistency ratio (CR) measures internal contradiction, and matrices with $CR \geq 0.10$ are customarily discarded [23]. Individual judgements may be synthesised into a group matrix by the element-wise geometric mean, which preserves reciprocity [24]. The method is attractive here because it yields an interpretable vector that a ranking engine can load at inference time.

D. Explainability in Recommendation

Explanations have been shown to support user trust and to help recipients judge whether a recommendation fits the stated need [25]. Post-hoc techniques such as LIME approximate a complex model locally [26], but the resulting narrative remains an approximation. A weighted linear fusion closes that gap: each factor's contribution is a term of the score, so the explanation is exact by construction.

E. Privacy and Regulation

Voice and health-related data fall under statute. The PDPA sets national obligations for personal data [27]; the GDPR defines lawful basis, purpose limitation, and data-subject rights [28]; HIPAA shapes health privacy and security practice [29]. In the present study these instruments function as design constraints on consent, minimisation, and auditability. Certification is not claimed.

F. Research Gap

The literature does not show that matching, speech, or mobile health are unstudied. It shows that they are studied apart. Roster tools keep calendars but rarely emit a scored, explained shortlist. NCD applications follow a condition but do not recommend caregivers. Health chatbots converse fluently but allow generated text to become the decision. Table I records the pattern. What remains absent, particularly for Sinhala and Tamil care requests, is an evaluated pipeline that converts speech into structured intent and then into an explainable, locally weighted, consent-aware ranking.

TABLE I
Capability Coverage of Existing System Classes

Capability	Roster	NCD app	Chatbot	Care Plus
Caregiver shortlist	Yes	No	Part	Yes
Voice intake (si / ta / en)	No	No	Part	Yes
Hybrid four-factor score	No	No	No	Yes
Consistency-checked weights	No	No	No	Yes
Emergency re-weighting	No	Part	No	Yes
Factor-level explanation	No	No	Part	Yes
Consent before voice AI	Part	Part	Part	Yes
LLM kept out of ranking	Yes	Yes	No	Yes

Part = partial or weak support. Classes are compared by feature because no shared public benchmark exists for this task.

III. Methodology

A. Design Science Research

Because the contribution is an evaluated artefact, the investigation follows Design Science Research (DSR) [30]. Peffers et al. [31] supplied the six-activity process; FEDS required judgement against explicit criteria rather than

demonstration alone [32]. A pragmatic stance combined quantitative checks with structured qualitative judgement [33]. Engineering proceeded in short Scrum-style increments inside that cycle [34].

B. Population, Sample, and Sampling Technique

The target population comprises healthcare stakeholders concerned with home-based chronic care in the Colombo District: patients and family members who seek care, professional caregivers and home health aides who supply it, clinicians who refer it, and agency administrators who organise it. No complete sampling frame of these groups exists. A probability draw was therefore neither feasible nor, for a design-science weight study, strictly required.

Stratified purposive sampling was adopted [33]. Strata were defined on three dimensions that later appear inside the matcher: stakeholder role (six categories), language used most for care (Sinhala, Tamil, English, bilingual), and Colombo zone band (city, inner suburb, outer suburb). Selection within strata was purposive rather than convenient, because every role and language that VEHMF must handle had to be present in the judgement set.

AHP can be applied to a small expert panel; an analytic floor of thirty consistent matrices is commonly treated as sufficient for a usable group vector. The planned sample was nevertheless set at 384 completed instruments so that, after consistency filtering, each stratum would still be represented and the aggregated eigenvector would not rest on a handful of matrices. That figure coincides with the conventional planning size for a 95% confidence interval at a 5% margin of error [35]. Coincidence is not identity: the study does not treat the resulting vector as a probability estimate of district preference. Of the 384 records obtained, family members accounted for 27.6% and patients for 22.1%; caregivers, home health aides, doctors or nurses, and agency administrators filled the remainder. Sinhala was the principal care language in 52.3% of records, bilingual use in 24.2%, English in 12.8%, and Tamil in 10.7%. Inner-suburban addresses formed the largest zone group.

The sample enters artefact development at one consequential point. An earlier prototype used a researcher-assigned coefficient set that placed nearly half the mass on clinical content. The group eigenvector from the 379 retained matrices (0.287, 0.231, 0.280, 0.202) replaced that assignment and is loaded at inference for ordinary matching. Sampling decisions therefore propagate into every non-urgent ranking. The vector remains a design input, not a population parameter for Colombo households.

C. Instrument, Consistency, and Aggregation

A structured instrument required pairwise comparison, on Saaty's scale, of four criteria aligned to the VEHMF factors: health compatibility, cultural and linguistic alignment, logistical (travel) feasibility, and trust and performance. Role, care language, and zone were recorded with the judgements. Each matrix was solved for its principal eigenvalue λ_{max} and eigenvector. The consistency index and ratio were computed as

$$CI = (\lambda_{max} - n) / (n - 1), \quad CR = CI / RI \qquad (1)$$

where $n = 4$ and $RI = 0.90$ is Saaty's random index for a four-by-four matrix. Matrices with $CR \geq 0.10$ were discarded. The remainder were combined by element-wise geometric mean [24] before the group vector was extracted.

D. Data for Ranking

Real caregiver-hiring logs are protected health-related data and were not collected. Seeded caregiver profiles—languages, Colombo coordinates, NCD specialties, and trust indicators—together with synthetic view, accept, and completion events, supported collaborative filtering and ranking tests. The practice is standard when privacy precludes the use of raw logs [36].

E. Evaluation Design

Three criteria were examined. Functionality was assessed by critical test cases derived from the requirements. Accuracy was assessed on two outputs: the consistency of the AHP vector, and the quality of the ranked list, using normalised discounted cumulative gain (NDCG) [37], with mean average precision and hit rate as complementary measures [36]. The acceptance rule required that hybrid ranking not score below a content-only baseline on the test relevance case. Usability was assessed by a seven-item structured walkthrough of the voice-to-shortlist path, informed by the System Usability Scale [38] but not reported as a population score. Performance was defined as a design target of 800 ms at the 95th percentile on a two-core, four-gigabyte CPU host.

F. Ethics

Instrument records were stored without names. Raw audio is processed only to extract intent and is not retained as a clinical archive. The dialogue policy forbids diagnosis. The final hiring decision remains with a person.

IV. Solution

A. System Architecture

Care Plus is realised as a lean modular monolith of five layers (Fig. 1). The presentation layer, built in React, captures microphone audio or typed text and renders the ranked shortlist. The API gateway uses the Django REST Framework for HTTP and Django Channels for a WebSocket that streams audio to the conversational layer. The cognitive layer, named Serah, employs the Gemini model to interpret the request and to produce a spoken reply. The decision layer hosts VEHMF. Users, caregivers, match runs, and consent records reside in PostgreSQL; Redis holds session traffic; an in-memory FAISS index supports content retrieval [15].

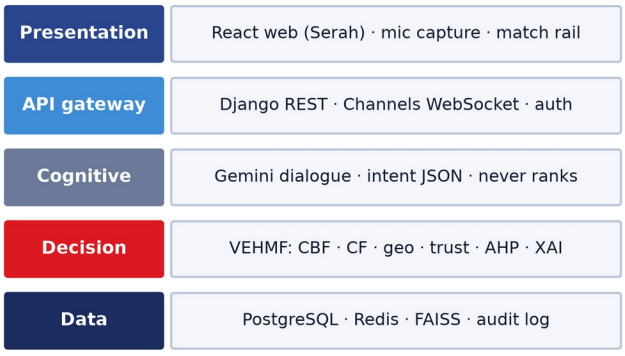


Fig. 1. Layered architecture of Care Plus. The language model is confined to the cognitive layer; ranking happens only in VEHMF.

B. From Speech to Structured Intent

A spoken or typed request is reduced to a constrained intent object that records condition, care level, preferred languages, urgency, specialty, and location. If a critical slot —condition or place—is empty, Serah asks a clarifying question and withholds ranking until the slot is filled. No audio or text is sent to an external model until the relevant consent scope has been granted. Only the structured intent, not the raw audio, is stored as the match object.

C. The VEHMF Ranking Model

For a care query q and an eligible caregiver c, four factor scores are computed. The content score s_{cbf} is the cosine similarity between the embedded intent and the caregiver's skill, condition, and language profile [14]. The collaborative score s_{cf} comes from implicit-feedback matrix factorisation over view, accept, and completion events [13]. The geographic score s_{geo} decays with the haversine distance between patient and caregiver [39], after a hard radius filter. The trust score s_{trust} aggregates reliability indicators such as completed visits and ratings. Each score is min–max normalised over the candidate set, and the final score is

$$S(q, c) = \alpha s_{cbf} + \beta s_{cf} + \gamma s_{geo} + \delta s_{trust} \quad (2)$$

where $\alpha + \beta + \gamma + \delta = 1$ and the weights are the group AHP vector. Candidates are sorted by S, and because S is a weighted sum, each factor's contribution can be shown directly to the user (Fig. 2). In the implementation, the language-and-culture weight β is applied to the second fusion slot; this mapping is stated explicitly so that β is not read as a survey preference for collaborative filtering.

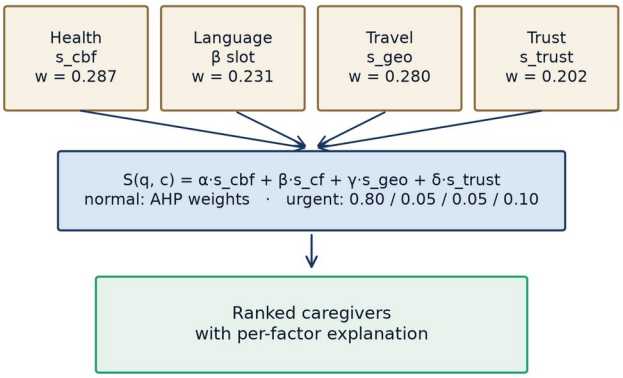


Fig. 2. VEHMF four-factor fusion. The factor bars shown to users are the same weighted terms that produce the rank.

D. Emergency Mode and Governance

When the intent is marked urgent, the engine re-ranks with a declared emergency vector of (0.80, 0.05, 0.05, 0.10), which prioritises medical fit. This vector is a design rule, not a second survey. If collaborative data are unavailable, β is set to zero and the remaining weights are renormalised. Accepting a caregiver places an optimistic hold on the schedule slot, and a conflict returns the next-best candidate instead of silently double-booking. Consent changes, match runs, and administrative actions are written to an audit log, and all transport is encrypted.

V. Evaluation

A. Functionality

Eight critical cases were derived from the requirements and implemented as automated tests. All eight passed (Table II). A pass indicates that the behaviour is implemented and protected by a test. It does not constitute clinical approval.

TABLE II
Critical Functionality Test Cases

ID	Behaviour tested	Result
F1	Intent extraction; missing slots trigger clarification	Pass
F2	Hybrid engine returns an ordered shortlist	Pass
F3	AHP weight vector sums to one	Pass
F4	Emergency vector (0.80, 0.05, 0.05, 0.10) applied	Pass
F5	Collaborative factor disabled; weights renormalised	Pass
F6	Missing consent blocks voice processing	Pass
F7	Schedule conflict returns next-best candidate	Pass
F8	Factor scores returned with each rank	Pass

B. AHP Weights

Of the 384 records, 379 matrices had CR < 0.10 and five were discarded. The mean individual CR of the retained set was 0.035. The group matrix had a principal eigenvalue of 4.000 and a CR of 0.00005, and was therefore accepted. Table III lists the weights. Health compatibility and travel feasibility are nearly equal; language and culture follow closely; trust accounts for about one fifth of the decision. The very low group CR is the expected consequence of geometric-mean aggregation of already consistent matrices; the mean individual CR is the more informative reliability figure.

TABLE III
Group AHP Weights from 379 Consistent Matrices

Criterion	VEHMF factor	Weight
Health compatibility	Content (α)	0.287
Language and culture	Second slot (β)	0.231
Travel feasibility	Geographic (γ)	0.280
Trust and performance	Trust (δ)	0.202
Mean individual CR	—	0.035
Group CR (λmax = 4.000)	—	0.00005

C. Ranking Quality

The ranking test compares a content-only ordering with the full hybrid ordering on a synthetic relevance list. Hybrid NDCG is required to be at least equal to content-only NDCG; the test passes. Adding the collaborative, geographic, and trust signals under AHP weights therefore does not degrade ranking quality on that case. A general percentage improvement is not reported, because no labelled set of real Colombo placements was available. A recommendation-accuracy figure would require such ground truth.

D. Usability

The seven-item walkthrough asked whether a user could state a need without a long form, recover when the need was incomplete, inspect the ranking order, inspect the four factors, notice an emergency re-ranking, retain control of the final decision, and use a preferred language. Six items were satisfied. The language item was partial: the pipeline accepts Sinhala, Tamil, and English, yet a native-speaker review of the interface text remains outstanding.

E. Achievement of Objectives

Table IV maps evidence to the four objectives. RO4 is recorded as partial because the 95th-percentile latency target has not been measured on a dedicated lean host.

TABLE IV
ACHIEVEMENT OF RESEARCH OBJECTIVES

Obj.	Evidence	Status
RO1	Gap analysis (Table I)	Achieved
RO2	VEHMF, AHP weights, F1–F8	Achieved
RO3	Consent gate, audit log, F6	Achieved (design)
RO4	Walkthrough 6/7; p95 not measured	Partial

VI. DISCUSSION

A. Interpretation of Findings

The findings demonstrate that caregiver-selection weights can be empirically quantified and systematically validated, rather than being based on unverified assumptions. Under the Colombo profile the travel coefficient is nearly indistinguishable from the clinical coefficient, and the language-and-culture coefficient is only slightly smaller. That distribution differs markedly from the earlier hand-set allocation, which had concentrated almost half of the decision on medical content.

Empirical comparison further indicates that a weighted hybrid constitutes a more defensible research object for this matching problem than either a solitary content score or an unconstrained language-model ranking. Each signal occupies a specified term in (2), and the acceptance rule precludes the hybrid from underperforming the simpler baseline on the test case. A related implication concerns the status of speech. Voice intake improves access only after the utterance has been reduced to structured intent. A transcript is not, by itself, a query. Until condition, language, urgency, and place are explicit, any subsequent ranker is interpolating; the clarification step is therefore part of the matching result rather than an incidental usability feature.

Institutional separation of conversation from ranking also addresses the hallucination risk reviewed in Section II [21]. Because the explanation is the weighted sum itself, it cannot drift from the order, unlike post-hoc approximations [26]. Embedding the consent gate in the pipeline likewise converts privacy from a policy statement into a behaviour that can be tested (F6).

B. Answers to the Research Questions

a) RQ1: Matching quality is improved when AI methods are combined inside a transparent weighted framework. VEHMF, loaded with the measured AHP vector, satisfied the rule of never scoring below content-only ranking on the test case.

b) RQ2: The spoken interface permits a request to be made without a form, and the clarify-then-match behaviour renders incomplete utterances usable. Engagement among less digitally confident users has not yet been measured on a recruited field sample.

c) RQ3: Admission of chronic-care intent allows condition and urgency to enter the score directly and, when required, to trigger the emergency vector, so the shortlist reflects the care need rather than proximity alone.

d) RQ4: Statutory compliance and operational efficiency are compatible when consent is checked before external processing, structured intent replaces raw audio, events are audited, and ranking executes locally on a CPU without a per-request model call.

C. Limitations

The weight vector derives from a stratified purposive instrument, not from a random sample of Colombo households, and cannot be read as a district parameter. Interaction histories are synthetic and cannot fully reproduce hiring behaviour. Recognition quality varies with noise, code-mixing, and microphone; geographic scores use distance rather than live travel time. The latency target and the native-speaker review remain outstanding. No clinical outcome was evaluated.

VII. CONCLUSION AND RECOMMENDATIONS

The study has developed and evaluated Care Plus, an AI-enhanced, voice-enabled personal healthcare assistant with integrated caregiver matching for chronic disease management in Sri Lanka. Spoken or typed requests in Sinhala, Tamil, or English are reduced to structured intent and ranked by VEHMF, a four-factor hybrid whose ordinary coefficients were elicited from 384 stakeholder records under stratified purposive sampling. What can be defended is the following: 379 consistent matrices with a group CR of 0.00005; eight of eight functionality cases passed; hybrid ranking that did not fall below a content-only baseline; and six of seven usability items satisfied. The language model supports dialogue and is excluded from rank determination, so every recommendation remains explainable and auditable, and a person retains the final decision.

Further work should close the remaining gaps in that order. The 95th-percentile match latency needs to be measured on a two-core, four-gigabyte host if RO4 is to be completed. A field AHP exercise with recruited Colombo residents would test whether the present vector holds beyond the designed instrument. An ethics-approved set of practitioner-graded queries would permit recommendation accuracy to be reported against real ground truth, and a native-speaker review of the Sinhala and Tamil interfaces would complete the accessibility claim that RQ2 only partly answers.

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